

Schools24 Development Cost Estimation

One-Time Development Cost (Simplified MVP)

Phase	Component	Services/Technologies Included	Cost (INR)	Cost (USD)
Phase 1	Architecture & Database Design	System architecture blueprint, PostgreSQL schema design, MongoDB collections, Redis data structures, API design documents	₹5,000	\$60
Phase 2	Backend Development (Core Services)	User Service, Academic Service, Financial Service, Notification Service, API Gateway setup, Authentication (JWT/OAuth2)	₹15,000	\$180
Phase 2	Frontend Development (Web App)	React dashboards for Student/Teacher/Admin, shadcn/ui components, Responsive layouts, State management (Redux/Zustand)	₹12,000	\$144
Phase 2	Mobile Apps (iOS & Android)	React Native apps for iOS/Android, Navigation, Offline sync, Camera/file upload, Push notifications setup	₹8,000	\$96
Phase 2	Android TV App	Android TV interface (Kotlin), Remote control navigation, Large-screen UI, Digital signage mode	₹3,000	\$36
Phase 3	API Integrations	Razorpay/Stripe payment gateway, Twilio/MSG91 SMS, SendGrid email, Google Maps API, Firebase Cloud Messaging	₹2,000	\$24
Phase 4	Local to Production Codebase	Complete testing suite, Docker containers, AWS/Azure cloud setup, monitoring systems, database optimization, security hardening	₹3,000	\$36
Phase 5	Documentation & Deployment	API documentation (Swagger), User guides, Admin manuals, Production deployment Ready, Training materials	₹2,000	\$24
TOTAL		Full microservices architecture with 5 platforms	₹50,000	\$600

What's Included in Each Phase

Phase 1: Architecture & Database Design (₹5,000)

Services & Deliverables:

- Complete system architecture blueprint with microservices layout
- Database schema design for 30+ tables:
 - **PostgreSQL:** Users, teachers, students, staff, classes, timetables, quizzes, quiz_questions, quiz_submissions, homework, homework_submissions, grades, attendance, study_materials, fee_structures, invoices, payments, leaderboard, student_usage, teacher_feedback, bus_routes, inventory, calendar_events
 - **MongoDB:** Questions collection, quiz_analytics, activity_logs, notifications_archive
 - **Redis:** Session keys, leaderboard sorted sets, cache structures, job queues
- API endpoint specifications (150+ endpoints across 8 services)
- Data flow diagrams showing inter-service communication
- Security architecture (authentication, authorization, encryption)
- ER diagrams with table relationships and foreign keys
- File storage bucket structure (S3 organization)

Phase 2: Backend Development (₹15,000)

8 Microservices Built:

1. **User Service** (Authentication & User Management):
 - User registration, login, logout, password reset
 - Role-based access control (Admin, Teacher, Student)
 - Profile management for all user types
 - Staff management (cleaners, drivers, custom roles)
 - JWT token generation and validation
2. **Academic Service** (Core Educational Features):
 - Timetable creation and management (daily schedules)
 - Quiz system: Create, schedule, auto-grade (MCQ support)
 - Homework: Upload assignments, track submissions, feedback
 - Grades management: FA1-4, SA1-2 assessments per subject
 - Attendance: Daily marking, statistics, reports
 - Leaderboard: Points-based ranking system
 - Study materials: Upload, categorize, distribute resources
 - Student monitoring: Dashboard usage, engagement tracking
3. **Financial Service** (Fee & Payment Management):
 - Fee structure configuration by class/grade
 - Invoice generation with itemized breakdown
 - Payment gateway integration (Razorpay/Stripe)
 - Payment tracking and receipt generation
 - Outstanding dues calculation and alerts
 - Payment history and transaction logs
 - Financial reports and analytics
4. **Notification Service** (Multi-Channel Communication):
 - Email service: SendGrid/AWS SES integration
 - SMS service: Twilio/MSG91 integration
 - Push notifications: Firebase Cloud Messaging
 - Notification templates (fee reminders, attendance alerts, exam notifications)
 - Delivery tracking and retry logic

- Queue management for async processing

5. **Monitoring Service** (Engagement Analytics):

- Student dashboard usage tracking
- Screen time monitoring and alerts
- Engagement score calculation
- Teacher activity tracking
- Performance metric aggregation

6. **Inventory Service** (Resource Management):

- Lab equipment inventory
- Library book catalog
- IT asset tracking
- Maintenance schedules
- Resource allocation system

7. **Bus Route Service** (Transportation Management):

- Route creation and optimization
- Driver assignment
- Student allocation to routes
- Google Maps API integration for visualization
- Route change notifications

8. **Reporting Service** (Report Generation):

- Student progress reports (academic, attendance, behavior)
- Teacher performance reports
- Financial reports (fee collection, outstanding dues)
- Custom report builder with filters
- PDF generation (Puppeteer/whhtmltopdf)
- Data export (CSV, Excel)

Technologies Used:

- Node.js/Express OR Python/Django for microservices
- PostgreSQL client (pg/psycopg2) for relational data
- MongoDB driver for document storage
- Redis client for caching and sessions
- WebSocket (Socket.io) for real-time features
- AWS SDK for S3 file uploads
- RESTful API architecture with proper error handling

Phase 2: Frontend Development (₹12,000)

3 Complete Web Applications:

1. **Student Dashboard** (10 pages):

- Dashboard: Subject progress (Science, Maths, Social, English, Hindi), assessment tracker (FA1-4, SA1-2), motivational quotes, quick actions
- Leaderboard: Class rankings with points, badges for top 3
- Quizzes: Take quizzes, view upcoming tests, results history
- Timetable: Daily schedule with subject-wise breakdown
- Calendar: School events, holidays, exam dates
- Materials: Download study materials by subject
- Fees: View fee structure, payment status, outstanding dues, payment history
- Attendance: Monthly attendance view, statistics, absence reasons
- Teacher Feedback: Per-assessment feedback with ratings and comments
- Reports: Academic progress, performance analytics

2. **Teacher Dashboard** (17 pages):

- Overview: Multi-class dashboard, student performance cards, engagement monitoring
- Student Monitoring: Track dashboard usage, screen time, engagement trends
- Quiz Scheduler: Create quizzes, add questions, schedule assessments
- Homework Uploader: Upload assignments, view submissions, provide feedback
- Materials Manager: Upload study materials, organize by class/subject
- Question Generator: AI-powered question creation tool
- Timetable Editor: Create/edit schedules for students and teachers
- Exam Scheduler: Schedule formal assessments, manage exam calendar
- Teach Module: Interactive teaching resources, lesson plans
- Attendance Upload: Mark daily attendance, bulk upload
- Leaderboard: View student rankings, performance metrics
- Teachers Leaderboard: Compare teacher performance
- Messages: Communication hub with students/parents
- Event Scheduler: Create and manage class events
- Student Timetable View: View student schedules
- Teachers Timetable: Personal schedule management
- Whiteboard: Teaching tools and resources

3. **Admin Panel** (15 pages):

- Dashboard: School-wide analytics, top performers, system health
- User Management: CRUD for all users, role assignment
- Student Details: Individual profiles, academic records, fee history
- Teacher Details: Teacher profiles, class assignments, subjects
- Staff Management: Non-teaching staff with salary tracking
- Class Management: Create classes, assign teachers, configure subjects

- Bus Route Management: Route planning, driver assignment, student allocation
- Student Timetable: School-wide student schedule management
- Teachers Timetable: Faculty schedule coordination
- Event Calendar: School events, holidays, exam schedules
- Teachers Leaderboard: Teacher performance rankings
- Students Leaderboard: School-wide student rankings
- Inventory: Lab equipment, library books, IT assets
- Fee Management: Configure fees, track payments, generate invoices
- Reports: Comprehensive analytics, data export

Common Features Across All Apps:

- Responsive design (mobile, tablet, desktop)
- Dark mode support
- Real-time notifications
- File upload with progress bars
- Data tables with sorting, filtering, pagination
- Charts and visualizations
- Form validation
- Toast notifications for user feedback

Technologies Used:

- React 18 with TypeScript for type safety
- Vite for lightning-fast build times
- shadcn/ui component library (40+ pre-built components)
- Tailwind CSS for utility-first styling
- React Router for navigation and nested routes
- React Context API for state management
- Axios for HTTP requests with interceptors
- Sonner for toast notifications
- Recharts for data visualization

Phase 2: Mobile Apps (₹8,000)

React Native Apps for iOS + Android (95% shared code):

Student Mobile App:

- All student web features optimized for mobile
- Native camera integration for document scanning
- Biometric authentication (fingerprint/Face ID)
- Offline mode with local data sync
- Push notifications with deep linking
- Pull-to-refresh for data updates
- Native file picker for homework uploads

Teacher Mobile App:

- All teacher web features optimized for mobile
- Quick attendance marking with camera
- On-the-go homework review and grading
- Mobile notifications for submissions
- Voice-to-text for feedback comments
- Native sharing for study materials

Common Mobile Features:

- Cross-platform support (95% code sharing)
- Native navigation with smooth transitions
- AsyncStorage for offline data
- Background sync when network available
- App icons, splash screens for both platforms
- Store optimization (App Store & Play Store listings)

Technologies Used:

- React Native CLI (not Expo for better native control)
- React Navigation 6 for routing
- AsyncStorage for local persistence
- React Native Camera for document scanning
- React Native Document Picker for file uploads
- Firebase Cloud Messaging for push notifications
- React Native Biometrics for authentication

Phase 2: Android TV App (₹3,000)

Large-Screen Classroom Display:

- Dashboard optimized for 1080p/4K displays
- Today's timetable with auto-refresh
- Live attendance tracker
- School announcements board
- Quiz results leaderboard
- Student performance metrics
- Remote control navigation (D-pad support)
- Screensaver mode with school branding
- Auto-launch on device boot

Technologies Used:

- Kotlin/Java with Android TV SDK
- Leanback library for TV-optimized UI
- Retrofit for API calls
- MVVM architecture pattern
- LiveData for reactive UI updates

Phase 3: API Integrations (₹2,000)

5 Third-Party Services Integrated:

- 1. Payment Gateway: Razorpay/Stripe**
 - Payment processing (UPI, cards, net banking)
 - Webhook handling for payment events
 - Refund processing
 - Invoice PDF generation
 - Payment link creation
- 2. SMS Gateway: Twilio/MSG91**
 - Bulk SMS sending (15,000/month included)
 - Template management
 - Delivery tracking
 - OTP sending for authentication
- 3. Email Service: SendGrid/AWS SES**
 - Transactional emails
 - HTML templates
 - Attachment support
 - Bounce handling
 - Delivery analytics
- 4. Maps API: Google Maps**
 - Bus route visualization
 - Geocoding addresses
 - Distance calculation
 - Route optimization
- 5. Push Notifications: Firebase Cloud Messaging**
 - iOS and Android push notifications
 - Topic-based subscriptions (per class/subject)
 - Scheduled notifications
 - Deep linking support

Phase 4: Local to Production Codebase (₹3,000)

Complete Production-Ready Infrastructure & Deployment Pipeline

1. Comprehensive Testing Suite (₹800)

- **Unit Tests:**
 - Frontend: Jest + React Testing Library (70%+ coverage)
 - Backend: Pytest with fixtures and mocks (80%+ coverage)
 - 500+ test cases covering all critical functions
- **Integration Tests:**
 - API endpoint testing (all 150+ endpoints)
 - Database transaction testing
 - File upload/download flow testing
 - Payment gateway integration testing
- **End-to-End Tests:**
 - Selenium/Cypress for critical user journeys
 - Student: Login → Take Quiz → Submit → View Results
 - Teacher: Create Quiz → Upload Material → Grade Homework
 - Admin: User Creation → Fee Setup → Report Generation
- **Performance Tests:**
 - Load testing with 500 concurrent users
 - Stress testing to find breaking points
 - API response time benchmarking (<300ms target)
- **Security Tests:**
 - SQL injection prevention testing
 - XSS vulnerability scanning
 - Authentication/authorization testing
 - Data encryption verification

2. Docker Containerization (₹500)

- **Dockerfiles** for all 8 microservices:
 - Optimized multi-stage builds (reduced image size by 60%)
 - Security scanning with Trivy

- Non-root user implementation
- **Docker Compose:**
 - Complete local development environment
 - Service orchestration (all services + databases)
 - Volume management for persistent data
 - Network configuration for inter-service communication
- **Container Registry:**
 - Push images to AWS ECR / Docker Hub
 - Image versioning and tagging strategy
 - Automated vulnerability scanning

3. Cloud Infrastructure Setup (₹1100)

- **AWS/Azure Complete Setup:**
 - **Compute:**
 - ECS/EKS cluster with 3 nodes
 - Auto-scaling policies (CPU/memory based)
 - Load balancer configuration (Application LB)
 - **Databases:**
 - RDS PostgreSQL (db.t3.medium, 100GB)
 - MongoDB Atlas M10 cluster
 - ElastiCache Redis (cache.t3.micro)
 - All with automated backups enabled
 - **Storage:**
 - S3 buckets with proper IAM policies
 - Lifecycle policies (archive after 6 months)
 - CloudFront CDN integration
 - CORS configuration for file uploads
 - **Networking:**
 - VPC with public/private subnets
 - Security groups with least privilege
 - NAT gateway for private subnet internet access
 - Route53 for DNS management
 - **Secrets Management:**
 - AWS Secrets Manager / Azure Key Vault
 - Encrypted environment variables
 - Automatic secret rotation

4. Monitoring & Observability (₹200)

- **CloudWatch Setup:**
 - Custom metrics for all services
 - Log aggregation from all containers
 - Alert rules for critical events:
 - Service downtime (>2 minutes)
 - Error rate >5%
 - Database connection pool exhausted
 - High CPU/memory usage (>80%)
 - Dashboard with key metrics visualization
- **Error Tracking:**
 - Sentry integration for exception monitoring
 - Stack traces with source mapping
 - User context in error reports
- **Uptime Monitoring:**
 - External health check every 1 minute
 - SMS/Email alerts on downtime
 - 99.5% uptime SLA tracking

5. Database Optimization (₹200)

- **Query Optimization:**
 - Index creation on frequently queried columns
 - Query performance analysis
 - N+1 query elimination
 - Connection pooling (PgBouncer/SQLAlchemy)
- **Backup Automation:**
 - Daily automated backups at 2 AM
 - 7-day point-in-time recovery
 - Cross-region backup replication
 - Backup restoration testing

- **Data Migration Scripts:**

- Version-controlled migrations (Alembic/Flyway)
- Forward and rollback scripts
- Data seeding for initial setup

6. Security Hardening (₹200)

- **SSL/TLS Configuration:**

- Let's Encrypt SSL certificates
- HTTPS enforcement (HTTP → HTTPS redirect)
- TLS 1.3 configuration
- HSTS headers enabled

- **Security Headers:**

- CSP (Content Security Policy)
- X-Frame-Options: DENY
- X-Content-Type-Options: nosniff
- Strict-Transport-Security

- **Rate Limiting:**

- 100 requests/minute per user
- Exponential backoff on auth failures
- DDoS protection via CloudFlare

- **Vulnerability Patching:**

- Dependency scanning (npm audit, safety)
- Automated security updates
- CVE monitoring and patching

Deliverables:

- ✓ 500+ automated tests (unit + integration + E2E)
- ✓ 8 Docker containers with optimized images
- ✓ Fully automated CI/CD pipeline (build → test → deploy)
- ✓ Complete AWS/Azure infrastructure (compute, database, storage, networking)
- ✓ Monitoring dashboard with 10+ custom metrics
- ✓ Automated backup system (daily backups, 7-day retention)
- ✓ Security hardening (SSL, rate limiting, vulnerability scanning)
- ✓ Database optimization (indexes, connection pooling, migrations)
- ✓ Load tested for 500 concurrent users
- ✓ Zero-downtime deployment capability

Technologies Used:

- **Testing:** Jest, Pytest, Cypress, Selenium, Postman, k6 (load testing)
- **Containerization:** Docker, Docker Compose, AWS ECR
- **CI/CD:** GitHub Actions, CodeDeploy
- **Cloud:** AWS (ECS/EKS, RDS, S3, CloudFront, Route53, CloudWatch)
- **Monitoring:** CloudWatch, Sentry, UptimeRobot
- **Security:** Let's Encrypt, Snyk, OWASP ZAP, Trivy
- **IaC:** Terraform (optional for reproducible infrastructure)

Phase 5: Documentation & Deployment (₹2,000)

Comprehensive Documentation:

API Documentation:

- Swagger/OpenAPI specs for all 150+ endpoints
- Interactive API testing interface
- Request/response examples
- Error code documentation

User Manuals:

- **Student Guide:** How to navigate dashboard, take quizzes, submit homework, view grades, check timetable, pay fees
- **Teacher Guide:** How to create quizzes, upload materials, grade homework, mark attendance, monitor students
- **Admin Guide:** System configuration, user management, fee structure setup, report generation, bulk operations

Technical Documentation:

- Architecture overview with diagrams
- Database schema with ER diagrams
- API integration guides
- Deployment runbook
- Troubleshooting guide
- Security best practices
- Backup and recovery procedures

Training Materials:

- Video tutorials (5-10 minutes each):
 - Student: Taking first quiz, submitting homework
 - Teacher: Creating quiz, uploading materials
 - Admin: Setting up fee structure, generating reports

- PowerPoint presentations for training sessions
- Quick reference cards for common tasks

Production Deployment Ready:

- Domain setup and SSL certificate
- Database migration scripts
- Initial data seeding (sample users, classes)
- Production environment configuration
- Load testing and performance tuning
- Security hardening
- Monitoring and alerting setup
- Backup verification

Post-Deployment Support:

- 1 week of bug fixes and stabilization
- Performance optimization
- User onboarding assistance
- Emergency hotfix support

Technology Stack Summary

Layer	Technologies
Frontend (Web)	React 18, TypeScript, Vite, Tailwind CSS, shadcn/ui, Zustand/Redux
Mobile	React Native, TypeScript, React Navigation, Firebase
Android TV	Kotlin, Android TV SDK, Leanback, Retrofit
Backend	Node.js/Express or Python/Django, JWT, WebSocket (Socket.io)
Databases	PostgreSQL (AWS RDS), MongoDB Atlas, Redis (ElastiCache)
File Storage	AWS S3, CloudFront CDN
Authentication	JWT tokens, bcrypt, OAuth2 (Google/Microsoft)
APIs	Razorpay/Stripe, Twilio/MSG91, SendGrid/AWS SES, Google Maps, Firebase
DevOps	Docker, Kubernetes/ECS, GitHub Actions, AWS/Azure
Monitoring	CloudWatch, ELK Stack (optional)
Testing	Jest, Pytest, Postman, React Testing Library

🏠 Conclusion

This architecture represents an **enterprise-grade, production-ready blueprint** for Schools24 that:

- ✔ **Scales** from 100 to 100,000 users
- ✔ **Secures** sensitive student and financial data
- ✔ **Performs** with < 300ms response times
- ✔ **Development Cost** ₹50,000 one-time (includes Android TV app)
- ✔ **Complies** with data protection regulations

Platform Coverage

Platform	Technology	Status
Web (Student/Teacher/Admin)	React	✔ Included
iOS Mobile	React Native	✔ Included
Android Mobile	React Native	✔ Included
Android TV	Kotlin (Android TV SDK)	✔ Included
Smart Board	Progressive Web App	✔ Included

Development Investment Summary

- One-Time Build Cost:** ₹50,000 (\$600 USD)
- Architecture & Database Design: ₹5,000
 - Backend (Core Services): ₹15,000
 - Web Dashboards: ₹12,000
 - Mobile Apps (iOS + Android): ₹8,000
 - Android TV App: ₹3,000
 - API Integrations: ₹2,000
 - Local to Production codebase: ₹3,000
 - Documentation & Deployment Ready: ₹2,000

Key Differentiators

Feature	Demo Architecture	Production Architecture (₹50K Budget)
Scalability	Single server, 100 users	Multi-server, 800-1200 users
Security	Basic login	MFA, encryption, RBAC
Performance	2-5 second load times	< 300ms API responses
Reliability	95% uptime	99.5% uptime
Data Safety	Manual backups	Automated, daily backups (7-day retention)

Document Version: 1.0
Last Updated: November 17, 2025
Contact: Architecture Team, Schools24
Status: Production Blueprint - Ready for Implementation

This production architecture is built to be secure, scalable, and resilient. Each service can be updated independently without taking the whole platform offline, which is exactly what an "MNC-level" client would expect.